

B.Tech. Examination-2024-25

Electronics and Communication Engineering Department

(Odd Semester Regular and Supplementary)

Course Title - Analog Electronics Course Code - (ECEUGPC01)

Full Marks : 80

Time-3 hrs.

(Answer any five taking at least one from each group, Symbols have their usual meaning)

Group A

1. (a) Explain the pinch-off and channel length modulation in n-channel MOSFET. Show that saturation region drain current expression in n-channel MOSFET with channel-length modulation is 4+5

$$I_D = \frac{1}{2} \mu_n C_{ox} \frac{W}{L} (V_{GS} - V_{TH})^2 (1 + \lambda V_{DS})$$

- (b) Determine the contact potential in an abrupt Si p-n junction with $N_a = 10^{18} \text{ cm}^{-3}$ in one side and $N_d = 5 \times 10^{15} \text{ cm}^{-3}$ on the other at room temperature. 5
- (c) Show that in BJT operating in active region $\beta = \frac{\alpha}{1-\alpha}$. 2

2. (a) Determine the Q-point (V_{CEQ}, I_{CQ}, I_{BQ}) in self bias Si-BJT circuit having $V_{CC} = 10V$, $R_1 = 5.6k\Omega$, $R_2 = 1k\Omega$, $R_C = 4.7k\Omega$, $R_E = 1k\Omega$ & $\beta = 10$. Draw the load line and Q-point of the circuit. 8

- (b) Explain thermal runaway in BJT. 2

- (c) Show that stability factor with respect to I_{CO} in self bias BJT circuit is 6

$$S_{I_{CO}} = \frac{\beta + 1}{1 + \beta \frac{R_E}{R_T + R_E}}$$

Group B

3. (a) Show that small signal voltage gain of the circuit shown in fig. 1 is 6

$$A_v = - (g_m R_C) \left(\frac{r_\pi}{R_B + r_\pi} \right)$$

- (b) Calculate the small-signal voltage gain of the BJT circuit shown in fig.1, with the parameters: $\beta = 100$, $V_{CC} = 12V$, $V_{BE} = 0.7V$, $R_C = 6k\Omega$, $R_B = 50k\Omega$, and $V_{BB} = 1.2V$. 10

4. (a) Show that the overall gain of the positive feedback amplifier is $A_{f+} = \frac{A}{1 - A\beta}$. 4

- (b) Show that negative-feedback increases the stability of the feedback amplifier. 3

- (c) Estimate the input impedance Z_{if} and output impedance Z_{of} of voltage-series feedback. 4+4

5. (a) Draw the common emitter configuration based voltage divider bias network for the ac response with the r_e model inserted between the appropriate terminals by including r_o and derive the expression of Z_i , Z_o and A_v . 10

- (b) A voltage-divider bias network based on CE configuration has the following specifications such as $V_{CC} = 16V$, $R_1 = 39k\Omega$, $R_2 = 4.7k\Omega$, $R_C = 3.9k\Omega$, $R_E = 1.2k\Omega$ & $\beta = 100$ and $C_1 = C_2 = 1\mu F$, $C_E = 10\mu F$. For this network determine the following parameters using the r_e model of transistor 6

- (i) r_e (ii) Z_i (with $r_o = \infty$) (iii) Z_o (with $r_o = \infty$) (iv) A_v

Group C

6. (a) State Barkhausen criteria for oscillation. 3

- (b) Determine the frequency of oscillation of the circuit shown in Fig. 2. 6

- (c) Derive the necessary formula. 7

7. (a) Identify the oscillator circuit and estimate the frequency of oscillation shown in Fig. 3. 1+6

- (b) What is the gain of the amplifier in sustain oscillation condition of the oscillator? 4

- (c) Explain why Clapp oscillator has more accurate and stable frequency of oscillation than Colpitts oscillator. 5

Group D

8. (a) Determine the output voltage for the log amplifier in Fig. 4. Assume $I_R = 50nA$. 5
- (b) Find the input-output relation of the circuit shown in Fig. 5. 5
- (c) Find out the output voltage V_O of the circuit shown in Fig. 6. 6
9. (a) Draw and explain the operation of pulse-generator using OPAMP with waveform diagram. 2+5
- (b) Derive the expression for the pulse duration of output waveform. 6
- (c) How the amplitude of the output can be adjusted with back-to-back Zener diodes. 3
10. (a) Calculate the frequency of oscillation and duty-cycle of astable multivibrator using IC-555 timer having parameters $R_A = 4.7k\Omega$, $R_B = 3.3k\Omega$ and $C = 0.01 \mu F$. 5+5
- (b) Derive the necessary formula. 6
11. (a) With help of appropriate circuit diagrams explain active low pass filter, active high pass filter, band pass and band stop filter. 8
- (b) Explain the operation of square wave generation process using Op-Amp with appropriate waveform diagram. 8

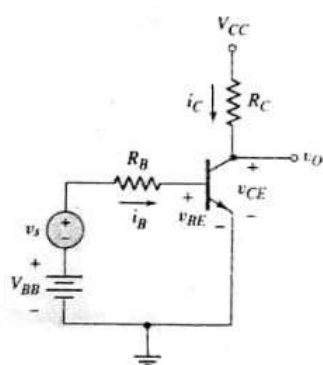


Fig. 1

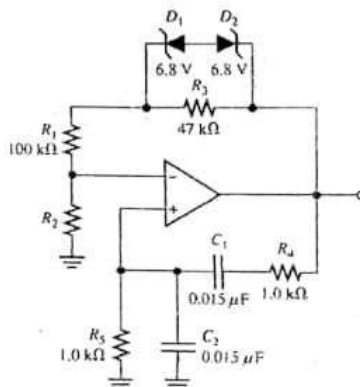


Fig. 2

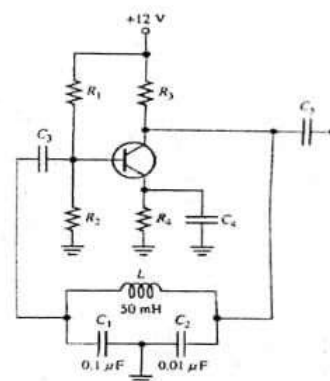


Fig. 3

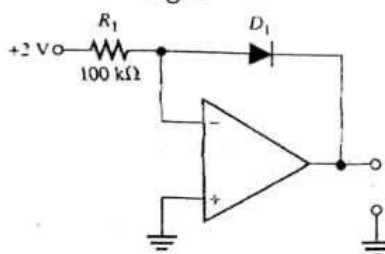


Fig. 4

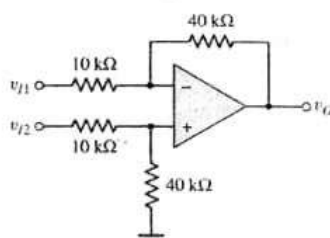


Fig. 5

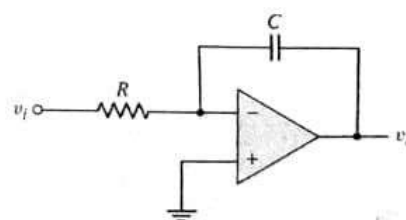


Fig. 6

Aliah University

End Term Examination Autumn Semester 2024
(For 2nd Year 3rd Semester of 4-Year B.Tech in ECE)

Subject: Data Structure & Algorithms

Paper Code: CSEUGOE01

Time: 3 hours

Full Marks: 80

Group - A (Answer any five)

2 x 5 = 10

1.

1. What is array? Why stack is called LIFO?
2. What is the pre-requisite of the Binary Search? Explain how sorted list ease the task of searching.
3. Explain how to represent doubly linked list with help of a diagram and an example?
4. If a problem has both iterative and recursive solution; what solution we should choose? Justify your answer.
5. Explain why the worst-case complexity of Quick sort is $O(n^2)$.

Group - B (Answer any five)

5 x 5 = 25

1. State the properties of an algorithm. Explain Space and Time Complexity of an algorithm with an example. 2 + 3
2. What is collision in hashing? Explain any two collision resolution techniques in hashing with example. 1 + 4
3. Explain the insertion and deletion operation in BST with an example (cover all possible cases). 5
4. Construct a tree for the given in-order and post-order traversals: In-Order - D, G, B, A, H, E, I, C, F; Post-Order: G, D, B, H, I, E, C, A. Why tree traversal algorithms are limited to three only? Justify. 3 + 2
5. What are the different types of Graphs available? How a Graph can be represented in computer memory? Explain all possible methods with an example. 1 + 1 + 3
6. What are the assumptions and advantages of Reverse Polish Notations? Give real life examples of Queue and Stack. 3 + 2
7. How a two-dimensional array is stored in computer memory? A 2-D array A[4:7, -1:3] requires 2 bytes of storage for each element. Find the address of element A[7,2] both in row major order and column major order if base address is 1000. 1 + 4

Group - C (Answer any three)

15 x 3 = 45

9. What are the advantages of balance tree over unbalanced tree? Explain with example.
10. Define the structure of a node in singly linked list in C. Compare array and linked list.
11. Evaluate the postfix expression using stack: 6, 2, 3, *, -, 3, 4, *, +, 3, 2, * - (explain each step). 2 + (1+3) + 5 + 4
12. Write the algorithm to merge two sorted arrays into a single array (sorted).
13. Explain how stack can be implemented using linked list.
14. Write an algorithm to find any particular element from given list of elements (sorted).
15. Explain the steps involved in insertion and deletion into a singly and doubly linked list.
16. Construct the AVL tree with the following elements: 25, 57, 38, 47, 12, 92, 86, 33, 1, 4, 3. 3 + 3 + 4 + 5
17. Write the algorithm / steps to insert a new element in Queue (Array implementation) to handle all possible case. Mention all the initial assumptions if any.
18. BFS/DFS traversal for a particular graph with same starting node is not fixed. Justify with suitable example.
19. Prove that the maximum number of edges of a graph of n vertices is $n * (n + 1) / 2$. List the differences between Graph and Tree in data structure.
20. Find out the Adjacency Matrix and Adjacency List representation of the following undirected Graph. Will these remain same if we add direction to all edges? Justify your answer.

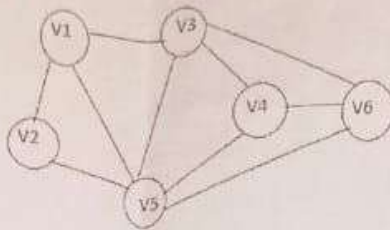


Fig. 1

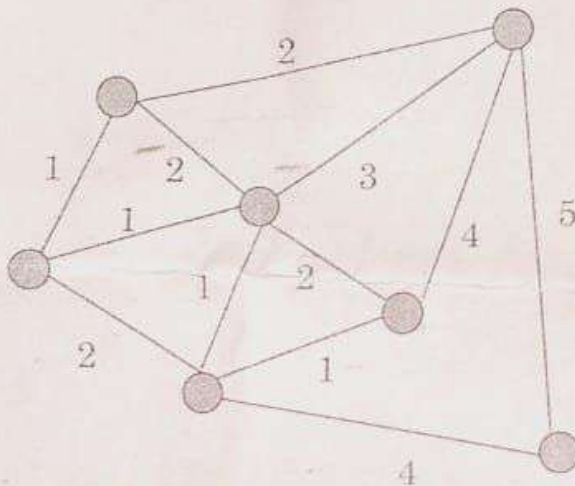
(c) What is balance factor in AVL tree?

$$3 + 3 + 4 + 4 + 1$$

12.

- (a) What will be the output of Quick sort algorithm after 3rd pass for the following sequence? 24 56 47 35 10 90 82 31
- (b) When a normal queue (Array implementation) will be full if the size of the queue is Max_SIZE? What will be the number of elements in the queue when it is full? Justify your answer.
- (c) Which algorithm will be best if the array is already sorted? What will be the complexity for this?
- (d) What are the differences between heap trees and binary search trees?
- (e) Define spanning tree. Find minimum spanning tree for the following graph:

$$3 + 3 + 2 + 2 + 5$$



B.Tech. Examination-2024-25
Department of Electronics and Communication Engineering
(Odd Semester Regular & Supplementary)

Course Title: Physics of Semiconductor Devices

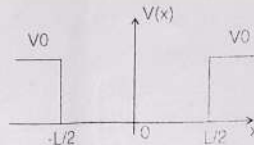
Course Code: ECEUGPC03

Full Marks: 80

Time: 3.00 Hrs

- Answer any five questions only.
- Answer all parts of a question in same place.
- Figures on the right hand side margin indicate full marks.
- Symbols have their usual meaning. Assume all necessary parameters as per their standard values.

1. Consider a quantum well with barrier height $V_0 = \infty$ and $V(-L/2 < x < L/2) = 0$ as shown below.



1. (a) Derive an expression for energy of the quantum states as function of its width. 8
- (b) Identify the ground state of the system. 2
- (c) Suppose there are two such quantum wells but with different widths, respectively L and $2L$. For a given maximum energy of a set of electrons, explain which well has capacity to store more electrons than the other. 6
2. (a) Derive expressions of effective density of states of conduction and valence bands in a bulk semiconductor. 6
- (b) For an intrinsic semiconductor, derive an expression for intrinsic carrier density in terms of effective density of states of conduction and valence bands. 6
- (c) How does the carrier density of an extrinsic semiconductor vary with doping density and temperature? 4
3. (a) What are a unit cell and the primitive cell? 4
- (b) Write the equations of the basis vectors of the inverse lattice in terms of the basis vectors of the direct lattice 6
- (c) For the two-dimensional hexagonal Graphene lattice, identify the unit cell in the lattice diagram. Write the expressions basis vectors in two-dimensional cartesian vector space. 6
4. (a) Derive the expressions of the electric field and the electrostatic potential in the space charge region of a linearly graded p-n junction. Assume all necessary parameters. 10
- (b) Design an abrupt silicon p-n junction in a p-type bulk region having $N_a = 10^{16} \text{ cm}^{-3}$, by compensating with a larger density of donor impurities (N_d). Specify the value of N_d to achieve a junction capacitance per unit area of 4.50 nF/cm^2 at a reverse bias of 5 V . Assume the $n_i = 1.5 \times 10^{10} \text{ cm}^{-3}$. 6
5. (a) Derive the forward bias current-voltage equation of an abrupt p-n junction. Assume all necessary parameters. Mention all assumptions. 10
- (b) Given for silicon at 300K the electron and hole diffusions constants are respectively $D_n = 25 \text{ cm}^2/\text{s}$ and $D_p = 10 \text{ cm}^2/\text{s}$, the excess minority carrier lifetimes $\tau_{n0} = \tau_{p0} = 0.5 \mu\text{s}$. The relative permittivity of silicon is 11.7 , and $n_i = 1.5 \times 10^{10} \text{ cm}^{-3}$. Design an abrupt silicon p-n junction, such that it conducts a total current density of 25 A/cm^2 at forward bias of 0.65 V . The 6

electron diffusion current should be four times the hole diffusion current. Specify the doping densities of p- and n- regions.

6. (a) Draw the device cross-section of a typical n-p-n bipolar junction transistor (BJT) in the context of an integrated circuit chip. 4
- (b) Derive the expressions of the steady state minority carrier distributions in the neutral base region of an n-p-n Bipolar Junction Transistor biased in forward active operating mode. Define all necessary quantities. 6
- (c) Explain the Early Effect and the punch-through condition in a BJT with reference to the minority carrier density in the neutral base region. 6
7. (a) Derive an expression of the base transport factor in an n-p-n BJT. 10
- (b) Design the base of an n-p-n BJT for the base transport factor of 0.9980. In the base region, the electron diffusion constant is $30 \text{ cm}^2 \cdot \text{s}^{-1}$ and the excess electron life time is $\tau_{n0} = 5 \times 10^{-7} \text{ s}$. Calculate the required base width. 6
8. (a) Draw the band diagrams of an enhancement mode NMOS transistor on silicon substrate along the transverse direction for the following conditions: 10
 - (i) $\phi_{\text{gate}} < \phi_{\text{sub}}$ at thermal equilibrium,
 - (ii) $\phi_{\text{gate}} < \phi_{\text{sub}}$ at the onset of weak inversion, and
 - (iii) $\phi_{\text{gate}} < \phi_{\text{sub}}$ at the onset of strong inversion.

The quantities ϕ_{gate} and ϕ_{sub} are respectively the work functions of the gate and the substrate. For each of these cases, derive expressions of the surface potential and the hole and electron density values at the Si-SiO₂ interface.
- (b) For an enhancement mode NMOS transistor in silicon substrate, having n⁺⁺-polysilicon gate and substrate doping of $N_A = 10^6$, calculate the gate-to-substrate contact potential at 300K. 4
- (c) What is channel length modulation effect? 2
9. (a) Derive an expression of the threshold voltage of an enhancement mode NMOS transistor. 5
- (b) Derive the current-voltage equation for an enhancement mode NMOS transistor for non-saturation and saturation operating conditions. Assume all necessary parameters. 6
- (c) Consider a bulk silicon NMOS fabrication process with $L_{\text{min}} = 1.25 \text{ } \mu\text{m}$, $\mu_n = 650 \text{ cm}^2/\text{V}\cdot\text{s}$, $V_{\text{Tn0}} = 0.65 \text{ V}$, $t_{\text{ox}} = 50 \text{ nm}$. The process supports two gate insulators, namely SiO₂ ($\epsilon_{\text{SiO}_2} = 3.9\epsilon_0$) and Si₃N₄ ($\epsilon_{\text{Si}_3\text{N}_4} = 7.5\epsilon_0$). For simplicity, assume that the choice of the gate dielectric does not change the threshold voltage. Design a minimum area NMOS transistor to drive a saturation drain current of $I_{\text{DSAT}} = 2 \text{ mA}$ at $V_{\text{GS}} = 3.65 \text{ V}$. Specify the gate insulator and the value of the gate width used in the design. 5

-End-

B.Tech. Examination-2024-25**Department of Electronics and Communication Engineering
(Odd Semester Regular & Supplementary)****Course Title: SIGNALS & NETWORKS****Course Code: ECE UGPC02****Full Marks: 80****Time: 3.00 Hrs**

- Answer all parts of a question in same place.
- Figures on the right hand side margin indicate full marks.
- Symbols have their usual meaning

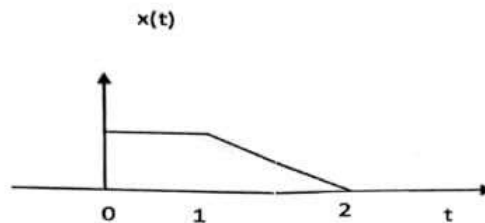
Answer Question No. 1 and any 5 from Question No. 2 to 9**Q. Questions****No.**Answer any *ten* from the following question: $10 \times 1 = 10$

1. (i) The integration of unit step signal is
☒ (a) ramp signal
 (b) impulse signal
 (c) exponential signal
 (d) parabolic signal
- (ii) If $x(n)$ is a discrete signal and $\delta(n)$ is a unit impulse signal then ,
 $x(n)\delta(n-n_0)$ is equal to,
 (a) $x(n)$ (b) $\delta(n)$ ☒ (c) $x(n_0)$ (d) $\delta(n_0)$
- (iii) Which of the following statements are true?
 (a) an LTI system is always stable system
 (b) an LTI system is stable only if the integral of its impulse response is finite
☒ (c) in a system, if the input is bounded then the output is also bounded
 (d) in a system, even if the input is bounded the output can be bounded
- (iv) Period of the signal $x(t) = 2\cos(2\pi t/3) + 3\cos(2\pi t/7)$ is
☒ (a) 2 (b) 7 (c) π (d) 21
- (v) Convolution between a finite signal and infinite impulse response gives out
☒ (a) Finite signal ☒ (b) Infinite signal (c) step signal (c) impulse signal
- (vi) If Fourier transform of $x(t)$ is $X(\Omega)$ then Fourier transform of $x'(t)$ is
 (a) $X(\Omega)/j\omega$ (b) $X(\Omega + j\omega)$ ☒ (c) $X(\Omega)/j\omega$ (d) $j\omega X(\Omega)$
- (vii) Which of the following signal is a periodic signal
 (a) $x(t) = Au(t)$ (b) $x(t) = Ae^{j\omega t}$ (c) $x(t) = Ae^{bt}$ ☒ (d) $x(t) = At$
- (viii) How can we define quarter wave symmetry?
 a) $x(t) = -x(t \pm T)$ and even
☒ b) $x(t) = x(t \pm T/2)$ and odd
 c) $x(t) = x(t \pm T)$ and odd
☒ d) $x(t) = -x(t \pm T/2)$ and even
- (ix) Fourier transform of $x(t) = u(t) - u(t - 2)$ gives out
 a) rect signal
 b) sinc signal
 c) gaussian signal
 d) unit step signal

- (x) Fourier series of a signal gives following output:
 a) Continuous-time and Periodic
 b) Discrete-time and Aperiodic
 c) Discrete-time and Periodic
 d) Continuous-time and Aperiodic
- (xi) Which of the following is a stable system?
 (a) $y(t) = tx(t)$ (b) $y(t) = t^2x(t)$ (c) $y(t) = \exp(t)x(t)$ (d) $y(t) = \exp(-t)x(t)$

2. (a) Given a signal $x(t)$ as follows

8



- Then Sketch (i) $x(t+1)$ (ii) $x(-t+1)$ (iii) $x(3t/2)$ (iv) $x(3t/2+1)$
- (b) Define the following :
 (i) Energy signal (ii) Power signal (iii) Non-causal signal (iv) Periodic signal

6

3. (a) Find the even and odd components of $x(t) = e^{j\omega t}$
 (b) Consider the system described by $y'(t) + 2y(t) = x(t) + x'(t)$
 Find the impulse response $h(t)$ of the system

6

8

4. (a) Find the Laplace transform of the following signal
 $\text{sgn}(t) = 1, t > 0$
 $= -1, t < 0$

5

- (b) $X(s) = 1/s(s+1)^2$, $\text{Re}(s) > -1$

5

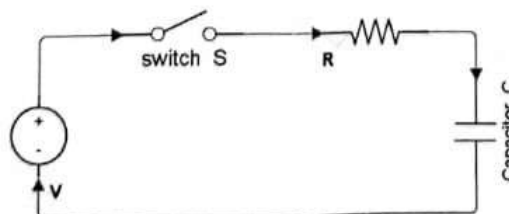
- (c) State and prove the Final value theorem of Laplace transform.

4

5. (a) Consider the RC circuit shown in the following Figure. The switch is closed at $t = 0$. Assume that there is an initial voltage on the capacitor and $v_c(0^-) = v_0$. Using Laplace transformation-

8

- (a) Find the current $i(t)$.
 (b) Find the voltage across the capacitor $v_c(t)$.



- (b) Discuss the ROC for Left sided, Right sided and Both sided signals and their stability issues.

6

6. (a) State and prove that in LTI system $y(t) = h(t) * x(t)$, $x(t)$ = Input signal, $y(t)$ = Output signal, $h(t)$ = Impulse response

6

(b) Find the convolution between the following signals-

5

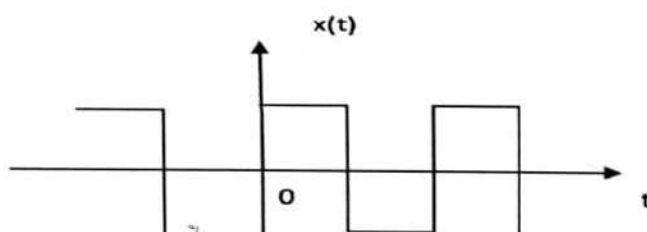
$$x(t) = u(t-1) - u(t-3) \text{ and } h(t) = u(t) - u(t-2)$$

(c) State the condition for linearity of LTI system.

3

7. (a) Find the Fourier series of the signal given by the following figure.

6



(b) Find the Fourier Transform of $x(t) = t \exp(at) u(t)$.

4

(c) Discuss the relation between Fourier transform and Laplace transform.

4

8. (a) Express the two port network in terms of open ckt impedance $[Z]$ parameter.

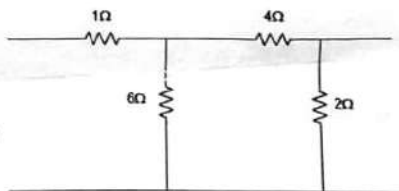
4

(b) What is symmetry and reciprocity condition of Z parameter, prove them.

4

(c) For the given network, find the short-circuit admittance $[Y]$ parameters. Also draw its equivalent circuit.

6



-End-

~~16 + 10 + 4 + 5 + 4~~
~~10 + 8 + 6 + 5 + 4 + 0 + 1 + 16~~

~~24~~
~~10~~
~~16~~
~~16~~

Aliah University
Theoretical Examination, 2024
UG section (Science & Engg/Autumn Semester)
Sub: Indian Constitution (UCCUGAU03)

Full Marks: 80

Time: 3hrs

A. Write any ten questions: (within 30 words each). 2x10=20

- ✓ 1. What is the significance of the Preamble of the Constitution of India?
- ✓ 2. Define 'Fundamental Rights'.
- ✓ 3. When was Abolition of the Untouchability Act enacted?
4. Discuss Article 21A of the Constitution of India.
5. Mention any two Socialist Principles of the Directive principles of State Policy.
- ✓ 6. Write two differences between Fundamental rights and Directive principle of state policy.
- ✓ 7. Discuss any two functions of the Speaker of the Lok Sabha.
- ✓ 8. What is impeachment? Has any President of India ever been impeached?
- ✓ 9. Mention the qualifications required to become a member of the Rajya Sabha.
- ✓ 10. Discuss the composition of the Upper House of the Indian Parliament.
- ✓ 11. Why the judgement in Kesavananda Bharti case considered one of the most celebrated judgements?
12. What is the significance of the 73rd Amendment to the Indian Constitution?
- ✓ 13. What are the primary functions of municipalities in India?

B. Write any four questions: (within 100 words each). 5x4=20

1. What are the differences between the Constitutional law and the Constitutionalism.
- ✓ 2. Write a short note on the writs of Habeas Corpus and Mandamus.
- ✓ 3. Describe the procedure for electing the President of India.
- ✓ 4. Mention Five Fundamental Duties enshrined in the Indian Constitution
- ✓ 5. Critically analyse the Right to Freedom of Religion as guaranteed under Articles 25 to 28.
6. Write a short note on the 74th Amendment Act

C. Write any four questions: (Within 350 words each). 10x4=40

- ✓ 1. Discuss the basic features of the Indian Constitution and explain how they reflect the vision of its framers.
2. Explain Article 19 of the Indian Constitution and highlight its reasonable restrictions.
3. "The Constitution of India is neither purely federal nor unitary but a combination of both." Do you agree with this view? Justify your answer.
- Or,
- ✓ Discuss the nature of Indian federalism'
4. Discuss the emergency powers of the President of India as provided in the Constitution. How do these powers ensure national security?
5. Explain the composition and functions of the Gram Panchayat in the Panchayati Raj system.
- ✓ 6. Explain the procedure for amending the Indian Constitution. How does it balance rigidity and flexibility?

Aliah University
End-Semester Examination - 2024 (Autumn)
Paper Code: MATUGBS03 Engineering Mathematics III
Semester – III (B.Tech. in CEN, MEN, ECE, EEN)

Full Marks – 80

Time – 3 Hours

Notations and symbols have their usual meaning.

Section A. Answer ALL questions. Each question carries one mark.

$1 \times 10 = 10$

1. The transformed equation of $3x + 4y = 5$ due to change of origin to the point $(3, -2)$ is
2. What is the angle between the lines $y^2 + xy - 2x^2 = 0$?
3. Give the condition that a given line $y = mx + c$ may touch the parabola $y^2 = 4ax$.
4. Find the equation of the chord of $x^2 + y^2 = 81$, which is bisected at $(-2, 3)$.
5. Find the distance of the plane $4x + 3y + 12z + 16 = 0$ from the point $(1, 2, 0)$.
6. The Laplace transform of the function $e^x \sin 6t$ is
7. Find the degree and order of $\frac{\partial^2 z}{\partial x^2} + \left(\frac{\partial^2 z}{\partial x \partial y}\right)^2 = 0$.
8. Classify the given partial differential equation $\frac{\partial^2 z}{\partial x^2} + 4\frac{\partial^2 z}{\partial x \partial y} + 4\frac{\partial^2 z}{\partial y^2} = 0$.
9. Write down the Charpit's auxiliary equation.
10. Find Fourier sine transform of $F(x) = e^{-ax}$, $a \geq 0$.

Section B. Answer any SEVEN questions. Each question carries 5 marks.

$5 \times 7 = 35$

11. The equation $3x^2 + 2xy + 3y^2 - 18x - 22y + 50 = 0$ is reduced to $4x^2 + 2y^2 = 1$ when referred to rectangular axes through the point $(2, 3)$. Find the inclination of the latter axes to the former.
12. Show that area of the triangle formed by the lines $ax^2 + 2hxy + by^2 = 0$ and $lx + my + n = 0$ is $\frac{n^2 \sqrt{h^2 - ab}}{am^2 - 2hlm + bl^2}$.
13. Show that the equation $8x^2 + 10xy + 3y^2 + 26x + 16y + 21 = 0$ represents a pair of straight lines and find the point of intersection and angle between them.
14. Reduce $11x^2 - 4xy + 14y^2 - 58x - 44y + 71 = 0$ to canonical form and state the type of the conic.
15. Show that four normals can be drawn to an ellipse through a given point and the feet of the normals lie on a rectangular hyperbola.

16. Find the locus of the poles of the normal chords of parabola $y^2 = 4ax$.
17. Two tangents drawn to $y^2 = 4ax$ meet at angle 45° . Find the locus of their point of intersection.
18. (a) Find the equation of plane parallel to x-axis and passing through the points $(1, -2, 4)$ and $(3, -4, 5)$.
 (b) Find the angle between the planes $2x + y + z = 6$ and $x - y + 2z = 3$. 3+2
19. (a) Find the shortest distance between the straight lines $\frac{x-3}{2} = \frac{y+15}{-7} = \frac{z-9}{5}$ and $\frac{x+1}{2} = \frac{y-1}{1} = \frac{z-9}{-3}$.
 (b) Find the equation of the straight line through the point $(3, 1, -6)$ and parallel to each of the planes $x + y + 2z - 9 = 0$ and $2x - 3y + z + 5 = 0$. 3+2
20. Find the equation of the sphere through the four points $(4, -1, 2)$, $(0, -2, 3)$, $(1, -5, -1)$ and $(2, 0, 1)$.

Section C. Answer any SEVEN questions. Each question carries 5 marks.

$5 \times 7 = 35$

21. By eliminating three arbitrary constants a, b, c from the family of ellipsoids

$$x^2/a^2 + y^2/b^2 + z^2/c^2 = 1, \text{ obtain}$$

$$zx \frac{\partial^2 z}{\partial x^2} + x \left(\frac{\partial z}{\partial x} \right)^2 - z \frac{\partial z}{\partial x} = 0; \quad zy \frac{\partial^2 z}{\partial y^2} + y \left(\frac{\partial z}{\partial y} \right)^2 - z \frac{\partial z}{\partial y} = 0; \quad z \frac{\partial^2 z}{\partial x \partial y} + pq = 0.$$

22. Find the partial differential equation arising from $\phi\left(\frac{z}{x}, \frac{y}{x}\right) = 0$.

23. Find the general solution of $z(x+y)p + z(x-y)q = x^2 + y^2$ by Lagrange's method.

24. Find the integral surface of the linear partial differential equation of $x(y^2 + z)p - y(x^2 + z)q = (x^2 - y^2)z$, which contains the straight line $x + y = 0, z = 1$.

25. Find the fourier series to represent $F(x) = e^{ax}$ in $(0, 2\pi)$.

26. Find the Fourier Sine Transform of

$$f(x) = e^{-ax}, a \geq 0.$$

Also show that

$$\int_0^\infty \frac{x \sin mx}{x^2 + a^2} dx = \frac{\pi}{2} e^{-ma}.$$

27. Eliminate the arbitrary function f and obtain partial differential equation for the following:

$$(a) \quad x + y + z = f(x^2 + y^2 + z^2) \quad z = f(x^2 - y^2).$$

28. Find the Laplace transform of (a) $t \cos^2 3t$ and (b) $e^{-4t} \sin 3t$.

29. Find the Fourier transform of

$$F(x) = \begin{cases} 1 - x^2, & |x| \leq 1 \\ 0, & |x| > 1. \end{cases}$$

30. Find the complete integral of $pxy + pq + qy = yz$ by Carpit's method.

End

2-1
2-2
2-3
2-4
2-5
2-6
2-7
2-8
2-9
2-10